# Pipeline Automation With GitHub

Using GitHub to Automate Testing

# In this video...

# What is automated testing?

- The use of specialized software (GitHub) to execute pre-written tests on application code
- To ensure that actual outcomes match with the expected results
- Helps identify bugs and ensure functionality
- In GitHub, the file is named and placed according to this structure:
  - root_directory/.github/workflows/main.yml

# Structure

- 3 sections:
  1. Workflow name
  2. Triggers:
     - When the tests run
  3. Jobs:
     - What to do

# Workflow name

```yaml
# 1. WORKFLOW NAME
name: <Any Name>
```

# Triggers

```yaml
# 2. TRIGGERS (WHEN IT RUNS)
on:
  push:
    branches: [<branch-1>, <branch-2>]

  pull_request:
    branches: [<branch-1>, <branch-2>]
```

# Jobs

```yaml
# 3. JOBS (WHAT IT DOES)
jobs:
  <job-id>:
    runs-on: <operating-system> # Ubuntu, Microsoft, etc

    steps:
      # Step 1: Get your code
      # Step 2: Set up environment
      # Step 3: Install dependencies
      # Step 4: Install testing tools
      # Step 5: Run tests

      # Each step must have a name:
      - name: <name>

      # Must be followed with a combination of:
      uses: <action-name>
      with:
          <key>: <value>
      run: <command>
```

# Combining

```yaml
# Workflow Name
name: Run CI/CD Tests

# Triggers
on:
  push:
    branches: [main, Section-2]
  pull_request:
    branches: [main, Section-2]

# Jobs
jobs:
  test:
    runs-on: ubuntu-latest

    steps:
    # Step 1: Get your code
    - name: Checkout repository
      uses: actions/checkout@v4

    # Step 2: Set up environment
    - name: Set up Python
      uses: actions/setup-python@v5
      with:
        python-version: "3.11.7"

    # Step 3: Install dependencies
    - name: Install dependancies
      run: python -m pip install --upgrade pip

    # Step 4: Install testing tools
    - name: Install pytest
      run: pip install pytest

    # Step 5: Run tests
    - name: Run tests
      run: pytest
```

# Live Demo

# Thank you for watching